

THE GENERALISED MULTIPLE CORRELATION MATRIX*

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1. Introduction

Let X be a random vector consisting of p variables $X' = (X_1, \dots, X_p)$. The interdependence between sets of variables of X are usually measured by means of correlation coefficients. The ordinary correlation coefficients ρ_{ij} ($i, j = 1, \dots, p, i \neq j$) measure the pairwise dependence of the variables of X and these correlations are usually displayed in the form of a matrix which is symmetric and called the correlation matrix. The multiple correlation coefficient measures the relationship between a variable of X and a linear combination of the remaining variables. The partial correlation coefficients measure the dependence between two variables of X after the effect of some (or all) of the other variables have been eliminated (see also Anderson (1958) for precise definitions).

If X is partitioned into two sets of components

$$(1) \quad X = \begin{pmatrix} X^{(1)} \\ X^{(2)} \end{pmatrix}; X^{(1)}: q \times 1, X^{(2)}: r \times 1,$$

then several "correlation coefficients" have already been defined to measure the relationship between the two sets. Well known are the canonical correlation coefficients, the vector correlation and the vector alienation coefficient (Hotelling (1936)). Less well known are the matrix correlation coefficient (Laurent (1957), Hooper (1959)), the trace correlation coefficient (Hooper (1959)) and the generalised multiple correlation matrix (Khatri (1964)).

If X is partitioned into three sets of components

$$X' = (X^{(1)}, X^{(2)}, X^{(3)}) (q \times 1, r \times 1, s \times 1)$$

then the relationship between variables in the sets may be measured by multiple partial correlation coefficients (Roy and Whittlesey (1952), Hooper (1962)), canonical partial correlation coefficients (Roy and Whittlesey (1952)), the partial trace correlation coefficient (Hooper (1962)) and the generalised multiple partial correlation matrix.

The purpose of this paper is to study the properties of the generalised multiple correlation matrix defined by Khatri (1964). As will be shown this matrix has the remarkable property that nearly all other "correlation coefficients" follow as special cases. This is similarly true for the generalised multiple partial cor-

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relation matrix in which nearly all partial correlations are embodied. The partial correlation matrix will be studied in a second paper.

Complex analogues of the correlations defined above can also be given for the case where X is a complex random vector (see for example Goodman (1963) Giri (1965), Khatri (1965)). In 6. of the present paper the complex analogue of the multiple correlation matrix will be defined.

2. The generalised multiple correlation matrix

Let the random vector $X: p \times 1$ be distributed according to a multivariate normal distribution $N(\mu, \Sigma)$ with mean μ and covariance matrix Σ . Let $X_{(1)}, \dots, X_{(N)} (N \geq p)$ be a random sample of N observations on X and let $A = \sum_{a=1}^N (X_{(a)} - \bar{X})(X_{(a)} - \bar{X})' = (a_{ij}); i, j = 1, \dots, p$, be the Wishart matrix. $\bar{X}$ and A/N are the maximum likelihood estimates of μ and Σ respectively.

Let X be partitioned into the two sets of components $X' = (X^{(1)'}, X^{(2)'})$ (as in (1)) where $X^{(1)}$ is $q \times 1$ and $X^{(2)}$ is $r \times 1$ with $q \leq r$, $r + q = p$. Partition Σ and A accordingly, that is

$$(2) \quad \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \quad A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$$

where Σ_{11} and A_{11} are $q \times q$, Σ_{12} and A_{12} are $q \times r$ and Σ_{22} and A_{22} are $r \times r$ matrices.

Since A_{11} is positive definite (A is positive definite) let $A_{11}^{1/2}$ be the positive definite square root of A_{11} . Any type of matrix square root may be considered. For example $A_{11}^{1/2}$ may be defined as $A_{11}^{1/2} = \Delta D^{1/2} \Delta'$ where Δ is an orthogonal matrix, $D = \text{diag}(a_1, \dots, a_q)$ and $a_1, \dots, a_q$ are the characteristic roots of $A_{11} = \Delta D \Delta'$, or A_{11} may be defined as $A_{11}^{1/2} = T$ where $A_{11} = T T'$. T is a nonsingular matrix and can be triangular. With the first definition $A_{11}^{1/2}$ is symmetric but this need not be the case with the second definition. In an expression such as $A_{11}^{1/2} G A_{11}^{1/2}$ we adopt the convention that the postmultiplier is $(A_{11}^{1/2})'$.

The generalised multiple correlation matrix

$$(3) \quad R = A_{11}^{-1/2} A_{12} A_{22}^{-1} A_{21} A_{11}^{-1/2}$$

was defined by Khatri (1964) as a measure of the correlation between the two sets of variables $X^{(1)}$ and $X^{(2)}$. The population generalised multiple correlation matrix can similarly be defined as the matrix

$$(4) \quad P = \Sigma_{11}^{-1/2} \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} \Sigma_{11}^{-1/2}$$

and it is obvious that R is the maximum likelihood estimate of P .

The matrix R has many interesting properties.

For $q = 1$, $r = 1$ and $p = 2$, (3) reduces to $R = a_{11}^{-1} a_{12} a_{22}^{-2} a_{21} = r_{12}^2$ the square of

the correlation coefficient between X_1 and X_2 . For $q = 1$ and $r = p-1$, $R = r_{1.2}^2 \dots r_{1.p}^2$ the square of the multiple correlation coefficient between X_1 and $(X_2, \dots, X_p)$ (see Anderson (1958)). It is because of these relationships between R and the correlation and multiple correlation coefficients that R was defined as the generalised multiple correlation matrix (Khatrı (1964)).

Suppose we wish to test whether the two sets $X^{(1)}$ and $X^{(2)}$ are independently distributed, that is we wish to test the hypothesis

$$(5) \quad H_0: \Sigma_{12} = 0.$$

The likelihood ratio criterion for testing H_0 is (Wilks (1932), Anderson (1958))

$$(6) \quad \lambda^{2/N} = W = \frac{|A_{11} - A_{12} A_{22}^{-1} A_{21}|}{|A_{11}|} \\ = \frac{|I - A_{11}^{-1/2} A_{12} A_{22}^{-1} A_{21} A_{11}^{-1/2}|}{|I - R|} = |I - R|$$

so that the likelihood ratio criterion is based on R .

In terms of Roy's union - intersection technique (Roy (1957)) the above hypothesis can also be stated as $H_0: \rho_1 = 0$ where ρ_1 is the first (largest canonical correlation coefficient between the two sets $X^{(1)}$ and $X^{(2)}$). The canonical correlation coefficients between the two sets are the square roots of the roots $\rho_1^2 > \rho_2^2 > \dots > \rho_q^2$ of the determinantal equation (Hotelling (1936), Anderson (1958)).

$$(7) \quad |\Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} - \rho^2 \Sigma_{11}| = 0$$

But (7) can be written as

$$(8) \quad 0 = |\Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} - \rho^2 \Sigma_{11}| = |\Sigma_{11}^{-1/2} \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} \Sigma_{11}^{-1/2} - \rho^2 I| = 0 \\ = |P - \rho^2 I|$$

so that the roots of the matrix P are the squares of the canonical correlation coefficients. The sample estimates of the canonical correlation coefficients are the square roots of the roots of

$$(9) \quad 0 = |A_{12} A_{22}^{-1} A_{21} - r^2 A_{11}| = |R - r^2 I|.$$

Roy's test is therefore based on the largest root of R . The test criteria proposed by Pillai (1955) for testing (5) are also based on R . The criteria are

$$(10) \quad V(q) = \sum_{i=1}^q r_i^2 = \text{tr } R \text{ and}$$

$$U(q) = \sum_{i=1}^q r_i^2 / (1 - r_i^2) = \text{tr } R (I - R)^{-1}$$

Furthermore the determinants of the matrices R and P

$$(11) |R| = |A_{12} A_{22}^{-1} A_{21}| / |A_{11}|, |P| = |\Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21}| / |\Sigma_{11}|$$

are the squares of the sample and population vector correlation coefficients.

It is also interesting to note that related matrix definitions of the correlation between two sets of variables have also been given by Laurent (1957) and Hooper (1959).

Laurent defined the $r \times q$ matrix

$$(12) Q = \Sigma_{22}^{-1/2} \Sigma_{21} \Sigma_{11}^{-1/2}$$

as the intuitively "natural" generalisation of the correlation between $X^{(1)}$ and $X^{(2)}$. The square roots involved in (12) are the non-symmetric (triangular) square roots. It is obvious that $P = Q'Q$. Laurent also derived the distribution of the sample estimate of Q .

Hooper's matrix generalisation of the correlation between sets of random variables is

$$(13) I - D = A_{11}^{-1} A_{12} A_{22}^{-1} A_{21}$$

so that the determinant $|I - D| = |R|$. Both $I - D$ and R have the same characteristic roots namely $r_1^2, \dots, r_q^2$ the squares of the canonical correlation coefficients. Hooper (1959) also defined the trace correlation coefficient as a measure of the relationship between $X^{(1)}$ and $X^{(2)}$. The trace correlation coefficient is $1/q \Sigma r_i^2 = 1/q \text{tr}(I - D) = 1/q \text{tr} R$.

Khatri's matrix generalisation of the correlation between sets of variables has an important advantage over the generalisations of Laurent and Hooper. R is symmetric while $I - D$ is non-symmetric and Q is non-square. The distributional properties of R seem to be easier to investigate than that of $I - D$ and the estimate of Q .

3. The distribution of the generalised multiple correlation matrix R

When $X^{(1)}$ and $X^{(2)}$ are independently distributed $\Sigma_{12} = 0$, $P = 0$ and the population canonical correlations are also zero. In this case the distribution of R is easy to derive.

Let

$$(14) B = A_{12} A_{22}^{-1} A_{21}$$

$$(15) E = A_{11} - A_{12} A_{22}^{-1} A_{21}$$

then

$$(16) R = (E + B)^{-1/2} B (E + B)^{-1/2}$$

From Theorem 4.3.2. of Anderson (1958), it follows that E has a Wishart distribution $W(\Sigma_{11}, n-r) (n = N-1)$ and B is independently distributed according to $W(\Sigma_{11}, r) (r \geq q)$. Thus R has a multivariate Beta Type 1 distribution $B_j(1/2r, 1/2n-1/2r)$ (Olkin & Rubin (1954), Troskie (1966)) with density function

$$(17) \beta(R) = (\Gamma_q(1/2n) / \Gamma_q(1/2r) / \Gamma_q(1/2n-1/2r)) |R|^{1/2(r-q-1)} |I-R|^{1/2(n-r-q-1)},$$

$$R > 0, I-R > 0$$

where

$$(18) \quad \Gamma_q(\ell/2n) = \Pi^{1/4q(q-1)} \Pi_{i=1}^q \Gamma(\ell/2(n-i+1))$$

is the multivariate Gamma function and $S > 0$ means that S is positive definite.

Since R has a multivariate Beta distribution the moments of the determinant $|R|$, the square of the vector correlation coefficient, are given by

$$(19) \quad E|R|^h = \Gamma_q(\ell/2n) \Gamma_q(\ell/2r+h) / \Gamma_q(\ell/2n+h) \Gamma_q(\ell/2r).$$

Similarly the moments of the likelihood ratio criterion are

$$(20) \quad EW^h = E|L-R|^h = \Gamma_q(\ell/2n) \Gamma_q(\ell/2n-\ell/2r+h) / \Gamma_q(\ell/2n+h) \Gamma_q(\ell/2n-\ell/2r)$$

For $q = 1$, $r = 1$ and $p = 2$ one can obtain from (17) and (19) the distribution and moments of $r_{1,2}^2$ the square of the correlation coefficient when the population correlation coefficient $\rho_{1,2}$ is zero. Similarly for $q = 1$ and $r = p - 1$ one obtains the distribution and moments of $r_{1,2}^2, \dots, p$ the (square of) multiple correlation coefficient when the population coefficient $\rho_{1,2}^2, \dots, p = 0$.

4. The non-central distribution of R

Suppose now that the sets $X^{(1)}$ and $X^{(2)}$ are correlated, that is $\Sigma_{1,2} \neq 0$, $P \neq 0$ and the population canonical correlations are different from zero. Let R be defined by

$$(21) \quad R = G^{-1/2} B G^{-1/2}, \quad G = B + E$$

where B and E are defined in (14) and (15) respectively.

Consider first the case where the second set of variables $X^{(2)}$ is kept fixed and let the observations on the second set be denoted by $X_{(a)}^{(2)} = x_{(a)}^{(2)}$, $a = 1, \dots, N$. From Theorem 4.3.2 of Anderson (1958) it follows that $E = A_{1,1} - A_{1,2} A_{2,2}^{-1} A_{2,1}$ has a conditional Wishart distribution $W(\Sigma_{1,1,2}, n-r)$ where $\Sigma_{1,1,2} = \Sigma_{1,1} - \Sigma_{1,2} \Sigma_{2,2}^{-1} \Sigma_{2,1}$. Since the distribution of E is independent of the fixed set of variables $X_{(a)}^{(2)} = x_{(a)}^{(2)}$ it is also the unconditional distribution. From the same theorem it follows that $B = A_{1,2} A_{2,2}^{-1} A_{2,1}$ is conditionally distributed according to a noncentral Wishart distribution $W(\Sigma_{1,1,2}, \Omega, r)$ (see Constantine (1963) or James (1964)) where Ω the matrix of noncentrality parameters, is given by

$$(22) \quad \Omega = \Sigma_{1,1,2}^{-1} \beta A_{2,2} \beta', \quad \beta = \Sigma_{1,2} \Sigma_{2,2}^{-1}.$$

The density (conditional) of B is

$$(23) \quad (\Gamma_q(\ell/2r))^{-1} (|\Sigma_{1,1,2}|)^{-1/2} \text{etr}(-1/2\Omega) \text{etr}(-1/2\Sigma_{1,1,2}^{-1}B) |B|^{\ell/2(r-q-1)} \\ {}_0F_1(\ell/2r; 1/4\Omega \Sigma_{1,1,2}^{-1}B), \quad B > 0$$

where $\text{etr}(S) = \exp(\text{tr } S)$.

The hypergeometric function of matrix argument is defined by (Constantine (1963))

$$(24) {}_sF_t(a_1, \dots, a_s; b_1, \dots, b_t; B) = \sum_{k=0}^{\infty} \sum_{\kappa} \frac{(a_1)_{\kappa} (a_2)_{\kappa} \dots (a_s)_{\kappa}}{(b_1)_{\kappa} (b_2)_{\kappa} \dots (b_t)_{\kappa}} \cdot \frac{C_{\kappa}(B)}{k!}$$

where $a_1, \dots, a_s, b_1, \dots, b_t$ are real or complex constants and the multivariate coefficient $(a)_{\kappa}$ is given by

$$(25) (a)_{\kappa} = \prod_{i=1}^q (a - \frac{1}{2}(i-1))_{\kappa_i}$$

where $(a)_{\kappa} = a(a+1)\dots(a+k-1) = \Gamma(a+k)/\Gamma(a)$. The partition κ of k is such that $\kappa = (k_1, \dots, k_q) (k_1 \geq k_2 \geq \dots \geq k_q \geq 0, k_1 + k_2 + \dots + k_q = k)$. The zonal polynomials $C_{\kappa}(B)$, being invariant under the orthogonal group, i.e. $C_{\kappa}(H'BH) = C_{\kappa}(B)$, $H \in O(q)$, are homogeneous symmetric polynomials in the characteristic roots of B . Zonal polynomials of products GB of symmetric matrices are defined as symmetric functions of the characteristic roots of GB . Since the roots of GB are the same as the roots of $G^{1/2}BG^{1/2}$, one may define $C_{\kappa}(GB) = C_{\kappa}(G^{1/2}BG^{1/2})$. For a discussion of the zonal polynomials the reader is referred to James (1960, 1961) and Constantine (1963).

It is obvious that the conditional density of B depends on the fixed set of variates $x^{(2)}$ since the noncentrality parameter $\Omega = \Sigma_{11}^{-1} {}_2\beta' A_{22} {}_2\beta$, where $A_{22} = \Sigma_a(x_{(a)}^{(2)} - \bar{x}^{(2)})(x_{(a)}^{(2)} - \bar{x}^{(2)})'$. Transforming from B and E to R and G by (21) with the Jacobian $J(B, E \rightarrow R, G) = |G|^{1/2} (p+1)$ one obtains the joint (conditional) distribution of R and G with density

$$(26) (\Gamma_q(1/2 n - 1/2 r) \Gamma_q(1/2 r) |2\Sigma_{11.2}|^{1/2} n)^{-1} \text{etr}(-1/2 \Omega) \text{etr}(-1/2 \Sigma_{11.2}^{-1} G) |G|^{1/2} (n-q-1) \cdot |R|^{1/2} (r-q-1) |I-R|^{1/2} (n-r-q-1) {}_0F_1(1/2 r; 1/4 \Omega \Sigma_{11.2}^{-1} G^{1/2} R G^{1/2}), G > 0,$$

$$R > 0, I-R > 0.$$

To obtain the unconditional distribution of R and G one must multiply (26) by the marginal density of $X_{(a)}^{(2)}$ ($a = 1, \dots, N$) and then integrate over $X_{(a)}^{(2)}$. However since (26) is a function of A_{22} , it is sufficient to integrate over A_{22} which has a Wishart distribution $W(\Sigma_{22}, n)$ with density

$$(27) (\Gamma_r(1/2 n) |2\Sigma_{22}|^{1/2} n)^{-1} |A_{22}|^{1/2} (n-r-1) \text{etr}(-1/2 \Sigma_{22}^{-1} A_{22}), A_{22} > 0.$$

Multiplying (26) by (27) and integrating term by term obtains the integral

$$(28) J = \int |A_{22}|^{1/2} (n-r-1) \text{etr}(-1/2 (\beta' \Sigma_{11.2}^{-1} \beta + \Sigma_{22}^{-1}) A_{22}) \cdot C_{\kappa}(1/4 \Sigma_{11.2}^{-1} \beta A_{22} \beta' \Sigma_{11.2}^{-1} G^{1/2} R G^{1/2}) dA_{22}.$$

To evaluate (28) make the transformation $A_{22} = \Sigma_{22}^{1/2} S \Sigma_{22}^{1/2}$ ($J(A_{22} \rightarrow S) = |\Sigma_{22}|^{1/2} (r+1)$) and then use the integral identity (Constantine (1963)),

$$(29) \int_{S > 0} \text{etr}(-RS) |S|^{t-1/2} (m+1) C_{\kappa}(ST) dS$$

$$= \Gamma_m(t)(t)_K C_K (R^{-1}T) |R|^{-t}, R > 0, S > 0,$$

where the integration in (29) is over the space of positive definite symmetric $m \times m$ matrices.

This yields

$$(30) \quad I = |\Sigma_{22}|^{1/2} n^{1/2} (I + \Lambda)^{-1/2} n \Gamma_r(1/2 n) (\frac{1}{2} n)_r \\ \cdot C_K (\frac{1}{2} (I + \Lambda)^{-1} \Sigma_{22}^{-1/2} \Sigma_{21} \Sigma_{11}^{-1} \Sigma_{12}^{-1/2} G^{1/2} R G^{1/2} \Sigma_{11}^{-1} \Sigma_{12} \Sigma_{22}^{-1/2})$$

where

$$(31) \quad \Lambda = \Sigma_{22}^{-1/2} \Sigma_{21} \Sigma_{11}^{-1} \Sigma_{12} \Sigma_{22}^{-1/2}.$$

The expression (30) can be simplified. If $\Sigma \Sigma^{-1} = I$, then $\Sigma_{11,2} = \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} = (\Sigma^{11})^{-1}$ and $\Sigma_{12} \Sigma_{22}^{-1} = -(\Sigma^{11})^{-1} \Sigma^{12}$ where

$$\Sigma^{-1} = \begin{bmatrix} \Sigma^{11} & \Sigma^{12} \\ \Sigma^{21} & \Sigma^{22} \end{bmatrix}$$

It is then easy to show that

$$(32) \quad |I_r + \Lambda|^{1/2 n} = |I_q - P|^{1/2 n}$$

where P is defined by (4) while the zonal polynomial in (30) reduces to $C_K (\frac{1}{2} \Sigma^{12} (\Sigma^{22})^{-1} \Sigma^{21} G^{1/2} R G^{1/2})$.

The joint density of G and R is then

$$(33) \quad f(R, G) = |I - P|^{1/2 n} \beta(R) (\Gamma_q(1/2 n) |2 \Sigma_{11,2}|^{1/2 n})^{-1} \\ \cdot |G|^{1/2 (n-q-1)} \text{etr}(-1/2 \Sigma_{11,2}^{-1} G) {}_1F_1(1/2 n; 1/2 r; 1/2 \Sigma^{12} (\Sigma^{22})^{-1} \Sigma^{21} G^{1/2} R G^{1/2})$$

where $\beta(R)$ is given by (17).

To do the actual integration over G which will yield an explicit expression for the density of R seems to be possible only for a very special case (i.e. the linear case.)* The factor $|I - P|^{1/2 n}$ appearing in (33) is very interesting. Intuitively one feels that if the integration with respect to G could be performed the resulting expression will probably be a function of P and R only. The following results support this view.

Theorem 1

The moment of order h of the determinant $|R|$, the square of the vector correlation coefficient, is given by

* After this research work was completed similar work was published by Srivastava (1968) (Ann. Math. Statist. **39**, 227-232) in which Srivastava gives an explicit expression for the distribution of R . His result, however, is incorrect.

$$(34) E|R|^h = (\Gamma_q(1/2 n) \Gamma_q(1/2 r+h) / \Gamma_q(1/2 n+h) \Gamma_q(1/2 r)) |I-P|^{1/2 n} \\ \cdot {}_2F_2(1/2 n, 1/2 n, 1/2 r+h; 1/2 n+h, 1/2 r; P)$$

where P is the population generalised multiple correlation matrix.

Proof

The following integral is needed for the proof (Constantine (1963)):

$$(35) \int |R|^{t-1/2(q+1)} |I-R|^u |I-R|^{-1/2(q+1)} C_K(UR) dR \\ = (\Gamma_q(t) \Gamma_q(u)(t)_K / \Gamma_q(t+u)(t+u)_K) C_K(U)$$

for $R > 0$ and $U > 0$.

If (33) is multiplied by $|R|^h$ and then integrated over $R > 0$ one obtains, after using (35) and (24), the expression

$$(36) (\Gamma_q(1/2 n) \Gamma_q(1/2 r+h) / \Gamma_q(1/2 r) \Gamma_q(1/2 n+h)) \\ \cdot |I-P|^{1/2 n} (\Gamma_q(1/2 n) |2\Sigma_{1,1}^{-1}|^{1/2 n})^{-1} |G|^{1/2(n-q-1)} \text{etr}(-1/2 \Sigma_{1,1}^{-1} G) \\ \cdot {}_2F_2(1/2 r+h, 1/2 n; 1/2 n+h, 1/2 r; 1/2 \Sigma^{1,2} (\Sigma^{2,2})^{-1} \Sigma^{2,1} G).$$

Finally integrating over G using (29), (24) and the fact that

$$(37) C_K(\Sigma_{1,1,2}^{-1} \Sigma^{1,2} (\Sigma^{2,2})^{-1} \Sigma^{2,1}) = C_K(P),$$

yields the result.

Theorem 2

The moment of order h of the likelihood ratio criterion $W = |I-R|$ is given by

$$(38) E W^h = E |I-R|^h = (\Gamma_q(1/2 n) \Gamma_q(1/2 n-1/2 r+h) / \Gamma_q(1/2 n+h) \Gamma_q(1/2 n-1/2 r)) |I-P|^{1/2 n} \\ \cdot {}_2F_1(1/2 n, 1/2 n; 1/2 n+h; P) *$$

where P is the population multiple correlation matrix.

Proof

Follows the line of the proof of Theorem 1.

From (33) one can also derive directly the joint distribution of the characteristic roots of R (that is the joint distribution of the squares of the canonical correlation coefficients). Since R is symmetric one can find an orthogonal matrix H such that

$$H'RH = \text{diag}(r_1^2, \dots, r_q^2), r_1^2 > r_2^2 > \dots > r_q^2$$

and H is defined in such a way to make the transformation unique (i.e. the elements of the first column of H are positive).

Using this transformation in (33) and then first integrating with respect to H

* This result has also recently been published by Sugiura and Fujikoshi (Ann. Math. Statist. 40 (1969), 942-952).

and then to G one obtains, after using well-known integration results proved in Constantine (1963) and James (1964), the joint distribution of $r_1^2, \dots, r_q^2$ as

$$(39) (\Pi^{1/2} q / \Gamma_q(1/2 q)) |I-P|^{1/2 n} \beta(R) a_q(R)$$

$${}_2F_1(1/2 n, 1/2 n; 1/2 r; P, R),$$

where $a_q(R) = \Pi_{i < j} (r_i^2 - r_j^2)$ and R and P are defined by (3) and (4), respectively. The joint distribution of $r_1^2, \dots, r_q^2$ was, of course, first derived by Constantine (1963) using a very similar method. However, (39) looks somewhat different from the density given by Constantine (1963) because of the very interesting appearance of the population generalized multiple correlation matrix P in the hypergeometric function. Of course the zonal polynomial involved $C_\kappa(P)$ is a polynomial in the characteristic roots $\rho_1^2, \dots, \rho_q^2$ of P (the squares of the population canonical correlations), so that (39) is exactly the density given by Constantine (1963).

5. The distributions of |R| and |I-R| for the linear case

The exact distribution of |R|, the square of the vector correlation coefficient, and of $W = |I-R|$ are extremely complicated. Explicit results can only be given for the linear case, that is when Σ_{12} is of rank one. This means that the multiple correlation matrix P is of rank one and hence has only the one root $\rho_1^2 \neq 0$.

The moments of |R| and |I-R| for the linear case can be found from (34) and (38) and are given by

$$(40) E|R|^h = (\Gamma_q(1/2 n) \Gamma_q(1/2 r + h) / \Gamma_q(1/2 n + h) \Gamma_q(1/2 r)) (1 - \rho_1^2)^{1/2 n}$$

$$\cdot {}_3F_2(1/2 n, 1/2 n; 1/2 r + h; 1/2 n + h, 1/2 r; \rho_1^2)$$

and

$$(41) E|I-R|^h = (\Gamma_q(1/2 n) \Gamma_q(1/2 n - 1/2 r + h) / \Gamma_q(1/2 n + h) \Gamma_q(1/2 n - 1/2 r)) (1 - \rho_1^2)^{1/2 n}$$

$$\cdot {}_2F_1(1/2 n, 1/2 n; 1/2 n + h; \rho_1^2)$$

since if P is of rank one then $|I-P| = 1 - \rho_1^2$.

Note that (40) gives the noncentral moments of the square of the correlation coefficient r^2 , when $q = 1$, $r = 1$ ($p = 2$), and the noncentral moments of $r_1^2, \dots, r_p^2$ when $q = 1$ and $r = p-1$.

From the moments (40) the exact distribution of R can be derived. Suppose that $Y_1, \dots, Y_q$ are independent Beta variables with density functions

$$(42) \frac{\Gamma(1/2(n-j+1))}{\Gamma(1/2(n-1/2r)) \Gamma(1/2(r-j+1))} y_j^{1/2(r-j+1)-1} (1-y_j)^{1/2(n-r)-1}, j = 2, \dots, q$$

and

$$(43) \frac{\Gamma(1/2 n)}{\Gamma(1/2 r) \Gamma(1/2(n-1/2r))} y_1^{1/2 r-1} (1-y_1)^{1/2(n-r)-1} (1-\rho_1^2)^{1/2 n} {}_2F_1(1/2 n, 1/2 n; 1/2 r; \rho_1^2 y_1)$$

Then it is easy to show that

$$\begin{aligned} E|R|^h &= \prod_{j=1}^q E(Y_j)^h \\ &= E(\prod_{j=1}^q Y_j)^h \end{aligned}$$

Hence the distribution of the (square of) vector correlation coefficient is the same as the product of q independent real Beta variables with density functions being given by (42) and (43).

Similarly can it be shown that the distribution of the likelihood ratio criterion $W = |I-R|$ is the same as the distribution of the product of q real independent Beta variables $Z_1, \dots, Z_q$ with density functions being given by

$$(44) \quad \frac{\Gamma(\frac{1}{2}(n-j+1))}{\Gamma(\frac{1}{2}(n-r-j+1))\Gamma(\frac{1}{2}r)} z_j^{\frac{1}{2}(n-r-j+1)-1} (1-z_j)^{\frac{1}{2}r-1} \text{ for } j = 2, \dots, q,$$

and

$$(45) \quad \frac{\Gamma(\frac{1}{2}n)}{\Gamma(\frac{1}{2}n-\frac{1}{2}r)\Gamma(\frac{1}{2}r)} \times z_1^{\frac{1}{2}(n-r)-1} (1-z_1)^{\frac{1}{2}r-1} (1-\rho_1^2)^{\frac{1}{2}n} {}_2F_1(\frac{1}{2}n, \frac{1}{2}n; \frac{1}{2}r; \rho_1^2(1-z_1))$$

The exact distribution of W for the noncentral linear case was first derived by Kshirsagar (1961).

6. The generalised multiple coherence matrix

In this section the complex analogues of the real variable results will be given. Derivations will not be given but these follow usually in the same way as for their real counterparts. For the benefit of the reader some preliminary definitions and results will, however, be given.

Let the complex random vector $Z = X + iY$ be distributed according to a complex normal distribution with density function (Wooding (1956), Goodman (1963)).

$$\pi^{-p} |\Sigma|^{-n} \exp(-\overline{(Z-\mu)}' \Sigma^{-1} (Z-\mu))$$

The distribution of Z will be denoted by $CN(\mu, \Sigma)$. The vector is a $p \times 1$ vector of complex parameters and the covariance matrix $\Sigma = E(Z-\mu)(\overline{Z-\mu})'$ is hermitian positive definite (hpd) (that is, $\Sigma = \overline{\Sigma}'$ where $\overline{\Sigma}'$ is the conjugate transpose of Σ).

Let $Z_{(1)}, \dots, Z_{(N)}$ be a random sample of size $N(>p)$ from Z and let $A = \Sigma(Z_{(a)} - \overline{Z})(\overline{Z_{(a)} - \overline{Z}})'$ be the complex Wishart matrix. A has the complex Wishart distribution $CW(\Sigma, n)$ ($n = N-1$) with density function (Goodman (1963)).

$$(\overline{\Gamma}_p(n) |\Sigma|^{-n})^{-1} |A|^{-n} \exp(-\text{etr}(-\Sigma^{-1}A)), A \text{ hpd},$$

where $\overline{\Gamma}_p(n) = \pi^{\frac{1}{2}p(p-1)} \prod_{i=1}^p \Gamma(n-i+1)$ is the complex multivariate Gamma function.

Let Z and Σ be partitioned into

$$Z = \begin{pmatrix} Z^{(1)} \\ Z^{(2)} \end{pmatrix} \quad \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

where $Z^{(1)}$ is $q \times 1$, $Z^{(2)}$ is $r \times 1$, Σ_{11} is $q \times q$, Σ_{12} is $q \times r$ and Σ_{22} is $r \times r$ ($q+r = p$, $q \leq r$).

If A is partitioned in the same way as Σ then the generalised multiple coherence matrix between the two sets $Z^{(1)}$ and $Z^{(2)}$ are defined as

$$R = A_{11}^{-1/2} A_{12} A_{22}^{-1} A_{21} \bar{A}_{11}^{-1/2}$$

The population multiple coherence matrix is

$$P = \Sigma_{11}^{-1/2} \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} \bar{\Sigma}_{11}^{-1/2}$$

It can be shown that R is the maximum likelihood estimate of P . For $q = 1$, $r = 1$ ($p = 2$), $R = r_{12}^2$ the sample coherence coefficient between Z_1 and Z_2 . For $q = 1$, $r = p-1$, $R = r_{12 \dots p}^2$ the sample multiple coherence between Z_1 and $(Z_2, \dots, Z_p)$.

The likelihood ratio criterion for testing the hypothesis of independence of the two complex sets $Z^{(1)}$ and $Z^{(2)}$ are again based on R . Using results of Giri (1965) one may show that the likelihood ratio criterion

$$\lambda^{1/N} = W = |I - R|$$

The roots of R and P are again the squares of the sample and population canonical correlation coefficients and $|R|$ may also be defined as the vector coherence coefficient between $Z^{(1)}$ and $Z^{(2)}$.

To find the distribution of R let R be defined by

$$R = (E + B)^{-1/2} B (\bar{E} + \bar{B})^{-1/2}, \quad E = A_{11} - A_{12} A_{22}^{-1} A_{21}, \quad B = A_{12} A_{22}^{-1} A_{21}.$$

The following results can be proved in the same way as are proved in Anderson (1958) for real variables.

Theorem 3

The conditional distribution of $E = A_{11} - A_{12} A_{22}^{-1} A_{21}$ given $Z^{(2)} = z^{(2)}$ is $CW(\Sigma_{11.2}, n-r)$. Since the conditional distribution is independent of $z^{(2)}$ it is also the unconditional distribution.

Theorem 4

The conditional distribution of $B = A_{12} A_{22}^{-1} A_{21}$ given $Z^{(2)} = z^{(2)}$ is the non-central complex Wishart distribution $CW(\Sigma_{11.2}, \Omega, r)(r > q)$ with matrix of non-centrality parameters given by

$$\Omega = \Sigma_{11.2}^{-1} (\beta A_{22})^{1/2} (\beta \bar{A}_{22}^{1/2})' = \Sigma_{11.2}^{-1} \beta A_{22} \bar{\beta}'$$

where $\beta = \Sigma_{12} \Sigma_{22}^{-1}$ and $A_{22} = \Sigma(z_{\alpha}^{(2)}, \bar{z}^{(2)}) / (z_{\alpha}^{(2)}, \bar{z}^{(2)})'$.

The density (conditional) of B is

$$(\bar{\Gamma}_q(n) |\Sigma_{11.2}|^n)^{-1} \text{etr}(-\Omega) \text{etr}(-\Sigma_{11.2}^{-1} B) |B|^{r-q} {}_0\bar{F}_1(r; \Sigma_{11.2}^{-1} B), B \text{ hpd.}$$

where ${}_0\bar{F}_1$ is the complex hypergeometric function of matrix argument defined by James (1964).

Using Theorems 3 and 4 the following results can be derived in the same way

as their real counterparts in 3, 4 and 5.

- (1) The density of the complex multiple coherence matrix R is

$$f(R) = (\bar{\Gamma}_q(n)/\Gamma_q(n-r)\Gamma_q(r))|I-P|^n \\ |R|^{r-q}|I-R|^{n-r-q}(\Gamma_q(n)|\Sigma_{1,2}^{-1}|^n)^{-1} \\ \int |G|^{n-q} \text{etr}(-\Sigma_{1,2}^{-1}G) \bar{F}_1(n; r; \Sigma^{1/2}(\Sigma^{2/2})^{-1}\Sigma^{2/2}G^{1/2}R\bar{G}^{1/2})dG;$$

G, r and $I-R$ hpd.

- (2) The moments of the vector coherence coefficient R are given by

$$E|R|^h = (\bar{\Gamma}_q(n)\bar{\Gamma}_q(r+h)/\bar{\Gamma}_q(n+h)\bar{\Gamma}_q(r))|I-P|^n {}_3\bar{F}_2(n, n, r+h; n+h, r; P)$$

- (3) The moments of the likelihood ratio criterion $W = |I-R|$ are

$$E W^h = E |I-R|^h = (\bar{\Gamma}_q(n)\bar{\Gamma}_q'(n-r+h)/\bar{\Gamma}_q(n+h)\bar{\Gamma}_q(n-r))|I-P|^n \\ \cdot {}_2\bar{F}_1(n, n; n+h; P)$$

- (4) The vector coherence coefficient R is distributed (linear case) as the product of q independent Beta variables $Y_1, \dots, Y_q$ with density functions $(\Gamma(n-j+1)/\Gamma(r-j+1)\Gamma(n-r))y_j^{r-j}(1-y_j)^{n-r-1}$, $j = 2, \dots, q$,

and

$$(\Gamma(n)/\Gamma(n-r)\Gamma(r))y_1^{r-1}(1-y_1)^{n-r-1}(1-\rho_1^2)^n {}_2F_1(n, n; \rho_1^2 y_1)$$

where ρ_1^2 is the only non-zero root of P .

- (5) The likelihood ratio criterion $W = |I-R|$ is distributed (linear case) as the product of q independent Beta variables $Z_1, \dots, Z_q$ with density functions $(\Gamma(n-j+1)/\Gamma(n-r-j+1)\Gamma(r))z_j^{n-r-j}(1-z_j)^{r-1}$, $j = 2, \dots, q$

and

$$(\Gamma(n)/\Gamma(n-r)\Gamma(r))z_1^{n-r-1}(1-z_1)^{r-1}(1-\rho_1^2)^n {}_2F_1(n, n; r; \rho_1^2(1-z_1)).$$

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